Percentages

* Percentage to fraction ($\div 100$)

$25\% = 25/100$

$60\% = 60/100$

$16 + (2/3)\% = \frac{50}{3 \times 100}$

$83 + (1/3)\% = \frac{250}{3 \times 100}$

* fraction to percentage ($\times 100$)

$3/4 = 3/4 \times 100 = 75\%$

$2/7 = 2/7 \times 100 = 28.57\%$

$\frac{1}{7} \rightarrow 14.28\%$

$\frac{2}{7} = 28.57\%$

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Q: A no. is inc. by 20%?

A: $100 + 20 = 120\%$

or

$\frac{20}{100} = \frac{1}{5}$ → change, → original

new = 1 + 5 = 6

Q: A no. is dec by 25%

$\frac{25}{100} = \frac{1}{4}$ → cha, → ori, → new = 4 − 1 = 3

Q: A no. is TT by 25%, then it b/m 600. The original no is?

A: $\frac{25}{100} \rightarrow \frac{1}{4} \rightarrow$ new = 1 + 4 = 5

$5x = 600$

$x = 120$

original = $4 \times 120 = 480$.

Q: A no. is dec by 20%, then it b/m 120. Original no?

A: $\frac{20}{100} = \frac{1}{5}$ new = 4 → 120

$x = 30$

Orig = 150.

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Q. No dt by $33+\frac{1}{3}\%$, it b/m 180. Orijial no?

A.

$\frac{100}{3\times100} = \frac{1}{3}$ → chg → ors. $3-1=2$

$2x = 180$

$x = 90$

Orijial = 270

Q. Total income of Yash of 2017, 2018, 2019 is 364 Rs. Every year salary ↑↑ by 20%. Income in 2017?

A.

2017 → 1

2018 → $1+\frac{1\times20}{100} = \frac{6}{5}$

2019 → $\frac{6}{5}+\frac{20}{100}\times\frac{6}{5} = \frac{36}{25}$

$1+\frac{6}{5}+\frac{36}{25} \Rightarrow \frac{91x}{25} = 364$

$x = \frac{364\times25}{91} = 100$

2017 → $x = 100$ Rs.

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▶ $x \cdot y =$ constant

$x \uparrow \quad y \downarrow \quad = K$

$y \uparrow \quad x \downarrow \quad = K$

Speed ↑ Time ↓ = distance const

of pop

$$\frac{r}{100 \pm r} \times 100$$

% I/p

% O/p

$$\frac{r}{1 \pm r} \times 100$$

fracⁿ I/p

% O/p

↑↑ → +

↓↓ → −

$$\frac{r}{1 \pm r} \times 1$$

fracⁿ +/p

fracⁿ O/p

Q: The price of petrol is ↑↑ by 10%, in order to keep the expend as constant, by what % consumption must reduced?

A. $\frac{10}{100 + 10} \times 100 \Rightarrow \frac{10}{110} \times 100 \Rightarrow \frac{100}{11} = 9.09\%$

Q: The price of cornion ↓↓ by 25%, in order to keep expenditure as const^n, by what % consumption must ↑↑ ?

A. $\frac{25}{100 - 25} \times 100 \Rightarrow \frac{1}{3} \times 100 = 33.33\%$

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Q. A is 25% less than B. By what % is B more than A?

A. $\frac{25}{100-25} \times 100 = 33.33\%$

Q. film indⁿ is 2/7 more rich than TV indⁿ. By what % is TV indⁿ less rich than film indⁿ?

A. $\frac{2/7}{1+2/7} \times 100 =$

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